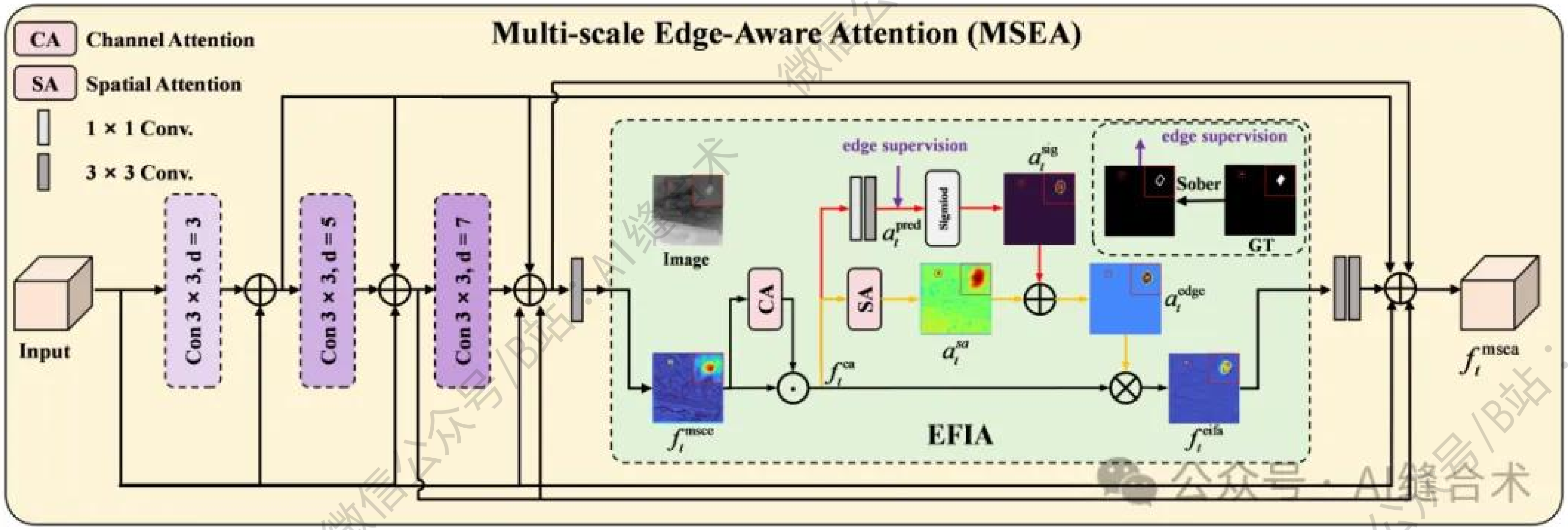
核心速览



论文题目： Edge-Semantic Synergy Network With Edge-Aware Attention for Infrared Small Target Detection

中文题目： 基于边缘感知注意力机制的边缘语义协同网络在红外小目标检测中的应用

论文链接：

https://ieeexplore.ieee.org/abstract/document/11314631

所属单位： 北京信息科技大学等

核心速览：

提出了一种具有边缘感知注意力的边缘语义协同网络（ESSNet），通过增强边缘信息和优化多级特征交互来提升红外小目标检测性能。

https://mp.weixin.qq.com/s/wAvRn44VZrBlKnzTxSW-ew

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  )
  (efia): EdgeFocusedIntegrationAttention(
    (ca): ChannelAttention(
      (avg_pool): AdaptiveAvgPool2d(output_size=1)
      (max_pool): AdaptiveMaxPool2d(output_size=1)
      (fc): Sequential(
        (0): Conv2d(32, 2, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (1): ReLU(inplace=True)
        (2): Conv2d(2, 32, kernel_size=(1, 1), stride=(1, 1), bias=False)
      )
    )
    (sa): SpatialAttention(
      (conv): Conv2d(2, 1, kernel_size=(7, 7), stride=(1, 1), padding=(3, 3), bias=False)
    )
    (edge_branch): Sequential(
      (0): Conv2d(32, 32, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (1): Conv2d(32, 1, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
    )
  )
  (refine): Sequential(
    (0): Conv2d(32, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
    (1): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (2): ReLU(inplace=True)
    (3): Conv2d(32, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
    (4): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  )
  (relu): ReLU(inplace=True)
)
input_tensor_shape  : torch.Size([2, 32, 256, 256])
增强后特征形状: torch.Size([2, 32, 256, 256])
边缘预测图形状: torch.Size([2, 1, 256, 256])

哔哩哔哩/微信公众号: AI缝合术, 独家整理!
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